In [2]:

```python
import pandas as pd
from sklearn.model_selection import train_test_split

# Load data
df = pd.read_csv("Heart.csv")

# Shape
print("Shape:", df.shape)

# Missing values
print("\nMissing Values:")
print(df.isnull().sum())

# Data types
print("\nData Types:")
print(df.dtypes)

# Zero values
print("\nZero Values:")
print((df == 0).sum())

# Mean age
print("\nMean Age:", df["Age"].mean())

# Select columns
data = df[["Age", "Sex", "ChestPain", "RestBP", "Chol"]]

print("\nSelected Data:")
print(data.head())

# Split data
train, test = train_test_split(
    data,
    test_size=0.25,
    random_state=42
)

print("\nTraining Data Shape:", train.shape)
print("Testing Data Shape:", test.shape)

print("\nTraining Data:")
print(train)

print("\nTesting Data:")
print(test)
```

```
Shape: (303, 15)

Missing Values:
Unnamed: 0    0
Age           0
Sex           0
ChestPain     0
RestBP        0
Chol          0
Fbs           0
RestECG       0
MaxHR         0
ExAng         0
Oldpeak       0
Slope         0
Ca            4
Thal          2
AHD           0
dtype: int64

Data Types:
Unnamed: 0      int64
Age             int64
Sex             int64
ChestPain      object
RestBP          int64
Chol            int64
Fbs             int64
RestECG         int64
MaxHR           int64
ExAng           int64
Oldpeak       float64
Slope           int64
Ca            float64
Thal           object
AHD            object
dtype: object

Zero Values:
Unnamed: 0      0
Age             0
Sex            97
ChestPain       0
RestBP          0
Chol            0
Fbs           258
RestECG       151
MaxHR           0
ExAng         204
Oldpeak        99
Slope           0
Ca            176
Thal            0
AHD             0
dtype: int64

Mean Age: 54.43894389438944
```

```
Selected Data:
   Age  Sex     ChestPain  RestBP  Chol
0   63    1       typical     145   233
1   67    1  asymptomatic     160   286
2   67    1  asymptomatic     120   229
3   37    1    nonanginal     130   250
4   41    0    nontypical     130   204

Training Data Shape: (227, 5)
Testing Data Shape: (76, 5)

Training Data:
     Age  Sex     ChestPain  RestBP  Chol
287   58    1    nontypical     125   220
282   55    0  asymptomatic     128   205
197   45    0  asymptomatic     138   236
158   60    1  asymptomatic     140   293
164   48    1    nonanginal     124   255
..   ...  ...           ...     ...   ...
188   54    1    nontypical     192   283
71    67    1  asymptomatic     125   254
106   59    1  asymptomatic     140   177
270   61    1  asymptomatic     140   207
102   57    0  asymptomatic     128   303

[227 rows x 5 columns]

Testing Data:
     Age  Sex     ChestPain  RestBP  Chol
179   53    1    nonanginal     130   246
228   54    1  asymptomatic     110   206
111   56    1  asymptomatic     125   249
246   58    1  asymptomatic     100   234
60    51    0  asymptomatic     130   305
..   ...  ...           ...     ...   ...
22    58    1    nontypical     120   284
258   70    1    nontypical     156   245
56    50    1    nonanginal     140   233
242   49    0  asymptomatic     130   269
114   62    0    nonanginal     130   263

[76 rows x 5 columns]
```